# Elouan Charbonnier

Cybersecurity Analyst - Malware Analysis, Reverse Engineering & CTI

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halftimeoflife.github.io | RootMe (2740 pts) | Driver's License B - French Nationality

## Profile

Master's graduate in Cryptology and Computer Security, specialising in malware analysis and reverse engineering. My internship at eShard and my published projects gave me solid expertise in static and dynamic analysis, evasion technique detection, and threat intelligence. I am targeting positions as malware analyst, reverse engineer, SOC analyst, or CTI analyst.

## Education

**University of Bordeaux** — **Bordeaux**
*Master's in Cryptology & Computer Security* — *2023–2025*

**University of Bordeaux** — **Bordeaux**
*Bachelor's in Mathematics & Computer Science* — *2020–2023*

## Certification

**Security Blue Team**
*BTL1 — BlueTeam Level 1 (Security Blue Team)* — *Obtained on July 23, 2026*
Modules covered: Phishing Analysis, Threat Intelligence, SIEM (Splunk), Digital Forensics, Incident Response.

## Professional Experience

**eShard** — **Pessac**
*Implementation, Detection and Bypass of Anti-Emulation Techniques on Windows* — *Mar–Aug 2025*
Reverse engineering internship on the analysis, implementation, and bypass of anti-VM/anti-emulation techniques in the context of eShard's TTD software (esReverse).

- Reverse-engineered real-world malware samples to identify anti-emulation markers (Ghidra, Binary Ninja, esReverse)
- Bypassed anti-VM protections via Windows API patching (WinDbg) and QEMU configuration tuning
- Implemented anti-VM techniques (CPUID, RDTSC, Windows API, ...) in x86 assembly and C
- Designed an automated binary generation pipeline embedding anti-VM techniques
- Automated detection of suspicious behaviours through TTD scripting (Python)
- Produced a technical article for eShard published on eShard's blog and an educational YouTube playlist

## Projects

**GhidraMAT**: Ghidra script framework (Python) for automated static detection of malicious TTPs (anti-VM, anti-debug, injection, persistence). Produces JSON/TXT reports and MITRE ATT&CK mappings.

**Malauto**: WinDbg extension automating dynamic analysis of packer behaviour on Windows.

**Wine APT29**: Full analysis of WINELOADER/GRAPELOADER (APT29, 2025): execution flow reconstruction, API deobfuscation, C2 identification with a proof of concept, technical report writing, and MITRE mapping.

**mispSK**: Collection of Python scripts (via PyMISP) automating operations on MISP instances.

**attackmap**: Python CLI tool generating interactive MITRE ATT&CK heatmaps from Navigator JSON layers.

## Publications

**2025**: Thesis — *"Detection of Anti-VM Protections in Malware"* [PDF]
**2025**: Article — *"Anti-Emulation Techniques on Windows"* [Article]
**2025**: Report — *"Linux Kernel Rootkit in 2025"* [PDF]

## Technical Skills

**Languages**: Python, C, x86/x64 Assembly
**Virtualisation**: QEMU, VMware, VirtualBox, FLARE-VM
**Static Analysis**: Ghidra, Binary Ninja
**Dynamic Analysis**: WinDbg, GDB, Reven TTD (eShard)
**CTI**: MITRE ATT&CK, YARA, VirusTotal, MISP
**Network**: Wireshark, Nmap
**SOC & IR**: Splunk, TheHive, DeepBlueCLI
**Forensics**: Volatility 3, Autopsy

## Languages

**French**: Native language
**English**: C1 Reading / B2 Listening (Linguaskill Certification)

## Interests

Hiking ▪ Bivouac